

current boards are not as efficient for medical device designing. The reason being "requirement of high-performance analogue-to-digital converter and digital-to-analogue converter for operating on physical signals received from the patient," he explains.

BITalino is an example of biomedical data-acquisition development board for the creation of projects using physiological sensors and tools. The board consists of Bluetooth, power, electromyography block, electrodermal activity block, electrocardiogram block, accelerometer, light emitting diode and light sensor. Powered by ATmega328 microcontroller, it can be configured with a sampling rate up to 1GHz and is capable of supporting six analogue inputs (four at 10-bit, two at 6-bit), four digital inputs and four digital outputs.

A board for music processing. Entertainment is an ever-increasing demand from the public. For a large part of the population, music has come as a safe haven. It is pretty easy when you have a music player in your phone and can plug in your headphones. This calls for attention to the quality of music, though. Engineers at the augmented instruments laboratory at C4DM, Queen Mary University of London, seem to have taken notice of the issue and have come up with a customised solution focussed towards musical applications.

Bela is an embedded system for real-time audio processing built on top of BeagleBone Black single-board computer. It features custom hardware and software environment with integrated audio processing and sensor connectivity in a single package. Built with

audio hardware projects in mind, the board does not stop there. It can also be used in projects requiring ultra-fast processing for multiple streams of data.

There is also the Axoloti development board. Axoloti Core is a circuit board with stereo audio in and output, audio analogue-to-digital and digital-to-analogue converters and a microcontroller suitable for digital audio processing. All connectors are on one side to enable a table-top device, rackmount, stompbox or something else. It can be connected to a PC or desktop and programmed via the patcher, and the music can be processed accordingly.

Moving from general-purpose to specific applications

Further, to ease the transition from a general-purpose board to an application-focused board, we have suitable integrated development environments (IDEs) to ease this process. "Moving directly from Arduino IDE to code composer is difficult at times," says Kamat.

This has been taken care of with Energia. It is an open source and community-driven integrated development environment and software framework. Kamat explains, "Usage of Energia is very similar to Arduino and, hence, is far easier to transition." However, code composer still remains the IDE of choice with veterans developing their projects on applications.

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The one that suits your requirement

Let us consider the very basic case of a musician who might be having a hard time processing music on a laptop. Now, the musician has the option to process stereo audio along with eight channels each of 16-bit analogue input/output and 16 channels of digital input/output, all of which are sampled together at audio rates. So is the case with other applications.

These solutions, being open source, bring out the tinkerer in most people. An added advantage with open source is easy access. Flodell says, "A customer can either get to production after having spent the minimum amount of time customising the initial design, or that the design is such that the customer can import it in whole or part into their own design."

Deshpande adds, "It gets easier to develop solutions with open source hardware becoming available." Since reference designs are easily available, "we can make changes in those to suit our applications as well," adds Kamat.

These advantages, along with smaller bills of material and time to market, make these boards very lucrative. With the boards being application-specific, there is a much higher chance to use these instead of adding shields to a general-purpose board and increasing its size. **EFY**